PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Chemical Kinetics & Radioactivity (Question Bank # 07)

MM : 75 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. For the reaction : $xA \longrightarrow yB$,

$$\log_{10}\left(-\frac{d[A]}{dt}\right) = \log_{10}\left(+\frac{d[B]}{dt}\right) + 0.3$$

If the value of $\log_{10} 5 = 0.7$, the value of x : y is :

(A) 2 : 1 (B) 1 : 2

(C) 3 : 10 (D) 5 : 2

2. For acid catalysed hydrolysis of ester rate law obtained is rate = k [ester] $[H^+]$,

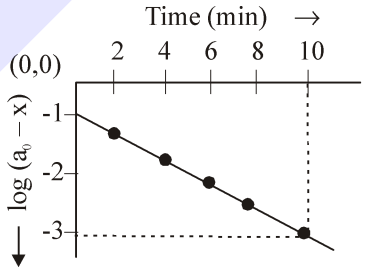
where k = 0.01 M^{-1} hr^{-1}. What is the half-life if the initial concentrations are 0.02 M for the ester and 0.05 M for the catalyzing acid.

(A) 1429 hours (B) 5000 hours

(C) 69.3 hours (D) 1386 hours

3. For the first order decomposition of $SO_2Cl_2(g)$, $SO_2Cl_2(g) \rightarrow SO_2(g) + Cl_2(g)$

a graph of $\log (a_0 - x)$ vs t is shown in figure. What is the rate constant (sec^{-1})?



(A) 0.2 (B) 4.6×10^{-1}

(C) 7.7×10^{-3} (D) 1.15×10^{-2}

4. If 0.01 % of a substance undergoing decomposition is consumed in millisecond when the concentration is 0.02 M and in 0.25 millisecond when the concentration is 0.04 M, what is the order of the reaction?

(A) 3 (B) 2 (C) 1 (D) zero

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5. A first order reaction : $A_{(g)} \rightarrow nB_{(g)}$, is started with pure A. The reaction takes place at constant volume and temperature . If the initial pressure of system was P_0 and the rate constant of reaction is 'K' min^{-1}, then the total pressure of the system at t = t min, is :

(A) $P_0.e^{-Kt}$ (B) $n.P_0\left(1-e^{-K1}\right)$

(C) $P_0\left[1+n.\left(1-e^{-Kt}\right)\right]$ (D) $P_0[n+(1-n).e^{-Kt}]$

6. C^{14} in a fossil of 'X' is presently 1/5 times of that in present atmosphere. Given that due to nuclear explosion after death of 'X' there was 25% increase in C^{14} content in atmosphere. The correct age of fossil is $x \times 10^4$ yrs, then find the value of x.

Given : $T_{1/2}$ for C^{14} = 6000 yrs.

(A) 1.20 (B) 1.6 (C) 2.4 (D) 1.8

7. Calculate the value of Activation energy at temperature of 300 K if the expression of rate constant as a function of temperature is given as $K = 30e^{\frac{-40}{T}}.e^{\frac{300}{T^2}}$.

(A) 40 R (B) 80 R (C) 300 R (D) 38 R

8. The radioactive decay $_{83}Bi^{211} \rightarrow {}_{81}Tl^{207}$, takes place in 100 L closed veseel at 27ºC. Starting with 2 moles of $_{83}Bi^{211}$ ($t_{1/2}$ = 130 sec), the pressure development in the vessel after 520 sec will be -

(A) 1.875 atm (B) 0.2155

(C) 0.4618 atm (D) 4.618 atm

(ii) One or more options correct Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

9. Which of the following is/are **correct** statement(s)?

(A) Stoichiometry of a reaction tells about the order of the elementary reactions.

(B) For a zero order reaction, rate and the rate constant are identical.

(C) A zero order reaction is controlled by factors other than concentration of reactants.

(D) A zero order reaction is always elementary reaction.

10. The activation energies of two reactions I and II are E_a and $2E_a$ respectively. If the temperature of the reacting systems is increased from T to T', predict which of the following alternative is/are correct ? (k represent rate constant)

(A) $k'_I / k_I > 1$ (B) $k'_{II} / k_{II} > 1$

(C) $k'_I / k_I < k'_{II} / k_{II}$ (D) $k'_I / k_I = 2k'_{II} / k_{II}$

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11. Select the correct option (s)

(A) Specific activity of U^{235} is greater than specific activity of U_3O_8 (All 'U' is U^{235})

(B) Activity of 1 mol U^{235} is greater than activity of 1mol U_3O_8 (All 'U' is U^{235})

(C) Decay constant of U^{235} is same whether it is pure U^{235} or U_3O_8 (All 'U' is U^{235})

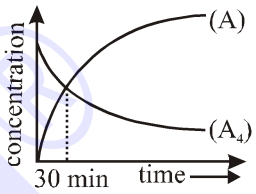
(D) Rate of radioactive disintegration increases with increase in temperature

12. Consider the following first order decomposition reaction,

$$A_4(g) \rightarrow 4A(g)$$

Which of the following statements regarding the reaction are correct?

[Given : log 2 = 0.30]



(A) At 30 min, only 20% reaction is complete

(B) $t_{1/2}$ of reaction is approximately 90 min.

(C) Rate of reaction decreases linearly with time .

(D) The time for intersection of two curves is independent of initial concentration of A_4.

13. Rate of radioactive decay for a radioactive element depends on :

(A) Amount of radioactive element

(B) Temperature

(C) Pressure

(D) Isotopic form of radioactive element

(iii) Paragraph Type

This section contains **2 paragraphs** , describing theory, experiment, data etc. **5 questions** relate to two paragraphs. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for question nos. 14 & 15

For a zero order chemical reaction,

$$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g)$$

rate of reaction = 0.1 atm/sec. Initially only NH_3 is present & Its pressure = 3atm.

14. Rate of disappearance of NH_3 will be

(A) 0.1 atm/sec. (B) 0.2 atm/sec.

(C) 0.05 atm/sec. (D) 0.04 atm/sec

15. Total pressure at t = 10 sec, will be

(A) 4 atm (B) 5 atm

(C) 4.5 atm (D) 3.5 atm

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Paragraph for question nos. 16 to 18

A substance 'X' dissociates by two parallel reactions with different ' Arrhenius constants (A_1 and A_2)" and different "Activation energies" $\left(E_{a_1} \text{ and } E_{a_2}\right)$ as shown.

$$X(g) \xrightarrow[E_{a_1} = 12 K\,cal/mol]{A_1 = 2\times10^{14} s^{-1}} Y(g) \quad ; \quad X(g) \xrightarrow[E_{a_2} = 9 K\,cal/mol]{A_2 = 2\times10^{14}\times e^{-2} s^{-1}} Z(g)$$

[**Given :** R = 2 Cal/mol-K]

16. Temperature at which net activation energy for dissociation of X will be 10.5 KCal/mol?

(A) 200 (B) 750 (C) 800 (D) 1000.

17. Which of the following statements are correct with respect to major product obtained in the above parallel decay:

(A) "Y" will be the major product at temperature 300 K.

(B) "Z" will be the major product at temperature 800 K.

(C) Both statements are correct

(D) None of the statements are correct

18. If the temperature is increased, then

(A) the rate of formation of Y will increase but the rate of formation of Z will decrease

(B) the rate of formation of both Y and Z will increase, but the percentage increase in rate will be higher for the formation of Y.

(C) the rate of formation of both Y and Z will increase but the percentage increase in rate will be higher for the formation of Z.

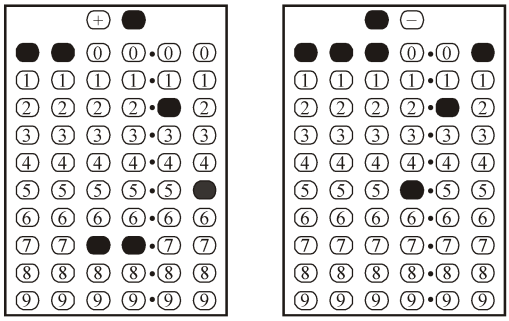
(D) the rate of formation of both Y and Z will increase, and the percentage increase in the rate will be same for both.

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SECTION-II: (Integer Second Decimal)

- This section contains **3** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

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1. A radioactive isotope disintegrates and emits 6 β and 7 α particles to form a stable product. Find the difference between atomic number of parent radioactive isotope and the stable product.

2. An optically active substance 'A' is decomposing into optically active substance 'B' and 'C' as

$$A \xrightarrow{K=10^{-3}\,min^{-1}} 2B + C$$

The specific rotations of A, B and C are + 40° , + 10° and –30° per mole, respectively. Initially 'A' and 'C' were present in 4 : 3 mole ratio. Close to how many hours the sample becomes optically inactive ?

[Given : $\ell n\frac{7}{5} = 0.34, \ell n\frac{20}{13} = 0.42$]

3. Kinetic data for hydrolysis of a 0.3 M solution of ethyl acetate in presence of HCl as catalyst is given in the following table. A constant volume of reaction mixture is taken at different time and titrated against standard alkali solution at different time.

Time / min	0	10	t = ∞
Vol. of Alkali / ml	25.00	28.00	40.00

It the rate law is given by rate = k[Ester]^1[H$^+$]^1, where H$^+$ is a catalyst, find the rate (in M/min) of ester hydrolysis in a solution which is 0.1 M each in ester and acid HCl (M/min)

Use: (ln 2 = 0.7, ln5 = 1.6)

If your answer is x × 10^{-y} then fill 'x' in OMR sheet.

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SECTION–IV : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and four statements (P, Q, R and S) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the Column :

	Column-I		**Column-II**
(A)	Molecularity of an elementary step reaction	(P)	Can never be –ive
(B)	Order of an elementary step	(Q)	Can be fractional
(C)	Order of a reactant in composite rate law	(R)	Is always natural number
(D)	Rate constant	(S)	Fixed value at constant temperature and reaction conditions

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ANSWER KEY

SECTION–I : (i) Only One option correct Type : 3(–1)

1.	Ans. (A)	2.	Ans. (D)	3.	Ans. (C)	4.	Ans. (A)
5.	Ans. (D)	6.	Ans. (A)	7.	Ans. (D)	8.	Ans. (C)

(ii) One or more options correct Type : 4(–1)

9.	Ans. (A,B,C)	10.	Ans. (A,B,C)	11.	Ans. (A,C)	12.	Ans. (A,B,D)
13.	Ans. (A, D)						

(iii) Paragraph Type: 3(0)

14.	Ans. (B)	15.	Ans. (B)	16.	Ans. (B)	17.	Ans. (D)
18.	Ans. (B)						

SECTION-II: (Integer Second Decimal) : 4(0)

1.	Ans. (8.00)	2.	Ans. (7.00)	3.	Ans. 4.00

SECTION–IV : Matrix-Match Type: 8(0)

1. Ans. (A)→(P,R,S) ; (B)→(P,R,S) ; (C)→(Q,S) ; (D)→(P,Q,S)

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